

MS4215: Lab 5

Mediation and Moderation: Exercise and Wellbeing

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Introduction

The dataset `exercise_wellbeing.csv` contains data from 200 participants in a study on physical activity and mental health. The variables are:

Variable	Description
<code>wellbeing</code>	Wellbeing score (1–10; higher = better wellbeing) — outcome
<code>exercise</code>	Exercise frequency (days per week, 1–7)
<code>stress</code>	Perceived stress level (1–10; higher = more stressed)
<code>support</code>	Social support score (1–5; higher = more support)

The lab has two parts. **Part A** examines whether stress *mediates* the relationship between exercise and wellbeing. **Part B** examines whether social support *moderates* that relationship.

Before attempting the questions, run the setup chunk below to load the required packages and data.

```
#install.packages("tidyverse")
library(tidyverse)

# In the line of code below, you will need to change the location to where you
# have saved the file
df <- read_csv("data/exercise_wellbeing.csv")
```

Questions - Part A: Mediation

(i) Explore the data

Use `summary()` to inspect all four variables. Then create scatterplots of:

- wellbeing vs exercise
- wellbeing vs stress
- stress vs exercise

Comment on the direction and strength of each relationship.

(ii) Three sequential regressions

Fit the three models required for the Baron & Kenny causal-steps approach:

- **Model 1:** `lm(wellbeing ~ exercise)`, for the total effect (path *c*)
- **Model 2:** `lm(stress ~ exercise)`, for path *a*
- **Model 3:** `lm(wellbeing ~ exercise + stress)`, for the direct effect (path *c*) and path *b*

Report the key coefficient and p-value from each model.

(iii) Interpret the mediation

Using `coef()`, extract paths *a* and *b* and compute the indirect effect $\hat{a} \times \hat{b}$.

- What proportion of the total effect operates through stress?
- Is the mediation partial or full? Justify your answer.
- Describe in plain language the mechanism by which exercise affects wellbeing.

Part B: Moderation

(iv) Fit the moderation model

Mean-centre `exercise` and `support` by subtracting their respective means. Then:

- Fit an additive model: `lm(wellbeing ~ exercise_c + support_c)`
- Fit an interaction model: `lm(wellbeing ~ exercise_c * support_c)`
- Use `anova()` to test whether the interaction term improves model fit significantly

(v) Interpret the interaction

Using the interaction model coefficients, write out the simple slope of `exercise_c` at:

- High social support: `support_c = +1 SD`
- Low social support: `support_c = -1 SD`

Plot the simple slopes using `expand.grid()` and `predict()`. What does the interaction tell us about when exercise is most beneficial for wellbeing?

(vi) Mediation vs moderation

In a short paragraph, explain the conceptual difference between mediation and moderation. Use exercise, stress, support, and wellbeing as your example.